STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 13: Oscillations

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Define period. [1]
2. Frequency-period relation? [1]
3. Restoring force in SHM? [1]
4. Where is speed maximum in SHM? [1]
5. Where is acceleration magnitude maximum? [1]
6. Spring angular frequency? [1]
7. Small-angle pendulum period? [1]
8. Total spring energy for amplitude A? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. Period .5s: frequency? [2]
10. m=.5kg,k=200N/m: period? [2]
11. For x=Acosomegat, acceleration? [2]
12. Pendulum length becomes 4l: period factor? [2]
13. Velocity at an extreme? [2]
14. Condition for pendulum SHM approximation? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. m=.5,k=200: angular frequency and period? [3]
16. A=.1m,omega=4: maximum speed and acceleration? [3]
17. Pendulum length 1m,g=9.8: period? [3]
18. Describe energy in one SHM cycle. [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. k=50N/m,A=.2m: total energy and U at x=.1m? [4]
20. Why does a real oscillator stop? [4]

End of paper - check working, units, signs and significant figures.